

# ABHINAV GUPTA

[agupta67@usc.edu](mailto:agupta67@usc.edu) | +1 (213) 414-9679 | [linkedin.com/in/abhinavg22](https://www.linkedin.com/in/abhinavg22) | [abhinavg-1010.github.io](https://github.com/abhinavg-1010)

CS Engineer with **2+ years** of hands-on experience in solving problems with **AI/ML and Data**. Proven track record of delivering high-quality work product, designing robust systems, and working cross-functionally in fast-paced environments.

## EDUCATION

### University of Southern California

August 2023 – May 2025

Master of Science – Computer Science

GPA: 3.83/4.0

Coursework – Machine Learning, Deep Learning, Multimodal, Natural Language Processing, Databases, Algorithms

### Vellore Institute of Technology, India

July 2019 – April 2023

Bachelor of Technology – Computer Science Engineering

GPA: 9.05/10.0

## SKILLS

**Languages:** Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), SQL, NoSQL, JavaScript, Flask (Streamlit), R, Java, C++

**ML Frameworks:** PyTorch, TensorFlow, LLM, Transformers, NLTK, HuggingFace, Docker, Generative AI, OpenCV

**Data:** Power BI, Tableau, SAS, Excel, MATLAB, Feature Engineering, Business Intelligence, Statistical Data Analytics

**Databases & Tools:** MySQL, Oracle, Salesforce, Firebase, AWS, GCP, Azure, Snowflake, Apache Spark, Hadoop, ArcGIS

## WORK EXPERIENCE

### Data Scientist

June 2024 – August 2024

*Esri*

*Redlands, CA, USA*

- Developed a **Forecasting Model** to analyze Esri Community health and forecast user behavior with **90% accuracy** using **Python Regression** prediction algorithm, enhancing quarterly metric tracking and annual business project roadmap planning
- Optimized **Retention Model** with Khoros API backend, advanced data visualization and **Statistical Data Analysis** to identify friction points for Esri Community members, thereby helping the team build actionable workflows to grow member retention
- Compiled dynamic ad-hoc data intelligence insight reports in **PowerBI** and created automated **data filtering pipelines** for faster reporting, resulting in improved business agility, better collaboration, data interpretation and reinforce team productivity

### Machine Learning Engineer

August 2023 – December 2023

*USC Information Sciences Institute*

*Los Angeles, CA, USA*

- Tackled issue of alarmingly rising aviation incursions through **ML** by developing a **BERT** multiclass classification model using **PyTorch** and **NLTK** that analyzes incident data and summary to identify and categorize leading human situational factors
- Sourced 470,000 reports which were extracted from a 1.4 million repository under carefully crafted filters from NASA's ASR System to apply targeted cluster indexing & find 6 key sub-classes reflecting fracture points to **avoid incursions by 21%**

### Data Scientist

September 2021 – March 2022

*Learn with Leaders*

*Delhi, India*

- Processed user engagement through data-driven analytics of 75,000 students by leveraging **SQL and Python** to re-design old, stale products, and draft adaptive marketing strategies, resulting in **40% boost** in organic website traffic
- Led the development of a dedicated **Performance Dashboard** in **Power BI** to identify key customer trends & interests, aiding content team design lucrative events & courses, building feedback adaptation, and contributing **30% increase** in customers

### Software Developer Engineer

October 2020 – August 2021

*Young Tycoons Business Challenge*

*Delhi, India*

- Co-founded a global entrepreneurship competition hosting 20,000 participants, **leading 10 interns** to develop a **full-stack** application portal hosted on **AWS** and a data dashboard for management, tracking, boosting outreach & scaling the business

## PROJECTS

### Multimodal SignToText Retrieval with Enhanced Facial Features ([Project Link](#))

January 2025 – May 2025

- Addressed absence of facial features in Sign Language Recognition using **PyTorch**, **MediaPipe** & **Transformer-based CLIP**
- Transformed **80hrs of video** to extract landmarks, design hand-pose-face features & process raw frame data for **Fusion** model
- Achieved **30% higher Recall** in SLR with explicit non-manual cues by training multimodal **video-to-text (V2T)** retrieval model

### Mitigating Catastrophic Forgetting in Continual Learning ([Project Link](#))

August 2024 – December 2024

- Solved challenge of catastrophic interference in class-incremental & task-agnostic continual learning using **PyTorch**, **Variational Autoencoders (VAEs)** and **Generative Adversarial Networks (GANs)** balancing stability and plasticity
- Applied Progressive **Dynamic Layer Expansion** & **Generative Memory Replay** resulting **6 times the accuracy** of Vanilla NN

## HONORS AND AWARDS

### Developer: Esri Intern Hackathon 2024 – 1<sup>st</sup> Place ([Project Link](#))

July 2024

- Competed among **15 top teams** & built EasyAisle, an indoor navigation system for grocery stores for helping doordashers
- Consolidated orders from multiple apps & calculated most optimal route with **ArcGIS Pro**, **Kotlin**, and **Firebase** in under 36hrs

### Co-Author: Analyzing Sentiment Analysis Using Supervised Learning ([DOI](#))

November 2022 – May 2023

- Conducted comparative study of sentiment analysis methods on 140,000 IMDB ratings, applying **7 classification algorithms**
- Reported effectiveness of majority probability Voting Classifier with best overall performance: 85.3% precision; 86.6% accuracy